

DFA Minimization

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The Big Picture

- Discover sets of equivalent states in the DFA
- Represent each such set with a single state

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Two states are equivalent if and only if:

- The set of paths leading to them are equivalent, *and*
- $\forall \alpha \in \Sigma$, transitions on α lead to equivalent states (DFA)

⇒ Must split a state that has α -transitions to distinct sets

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A partition P of Q

- A collection of sets P s.t. each $q \in Q$ is in exactly one $p_i \in P$
- The algorithm iteratively constructs partitions of the DFA's states

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Details of the algorithm

- Group states into maximally sized initial sets, *optimistically*
- Iteratively subdivide those sets, based on transition graph
- States that remain grouped together are equivalent

Initial partition, P_0 , has two sets: $\{F\}$ & $\{Q-F\}$ $D = (Q, \Sigma, \delta, q_0, F)$
final states others

Iterative Refinement: Splitting a set ("partitioning a set by a")

- Assume q_a & $q_b \in p_i$, and $\delta(q_a, \underline{a}) = q_x$, & $\delta(q_b, \underline{a}) = q_y$
- If q_x & q_y are not in the same set p_j , then p_i must be split
 - q_a has transition on a , q_b does not $\Rightarrow \underline{a}$ splits p_i
- One state in the final DFA cannot have two transitions on a

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The algorithm

```
T ← { F, {Q-F}}
P ← {}
while ( P ≠ T)
  P ← T
  T ← {}
  for each set  $p_i \in P$ 
    T ← T  $\cup$  Split( $p_i$ )

Split(S)
  for each  $c \in \Sigma$ 
    if  $c$  splits  $S$  into  $s_1$  &  $s_2$ 
      then return { $s_1, s_2$ }
  return S
```

Why does this work?

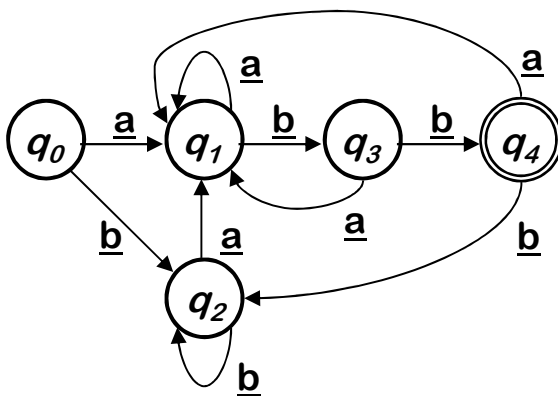
- Partition $P \in 2^Q$
- Start off with 2 subsets of Q : $\{F\}$ and $\{Q-F\}$
- The *while* loop takes $P_i \rightarrow P_{i+1}$ by splitting 1 or more sets
- P_{i+1} is at least one step closer to the partition with $|Q|$ sets
- Maximum of $|Q|$ splits

Note that

- Partitions are never combined
- Initial partition ensures that final states remain final states

A Detailed Example (DFA Minimization)

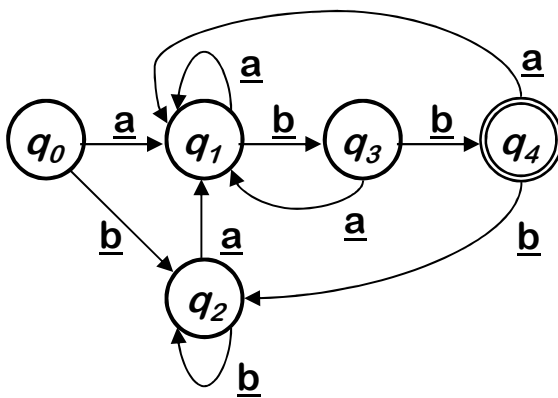
	<i>Current Partition</i>	<i>Worklist</i>	<i>s</i>	<i>Split on <u>a</u></i>	<i>Split on <u>b</u></i>
P_0	$\{q_4\} \{q_0, q_1, q_2, q_3\}$	$\{q_4\} \{q_0, q_1, q_2, q_3\}$			



For the record, example was right in 1999, broken in 2000

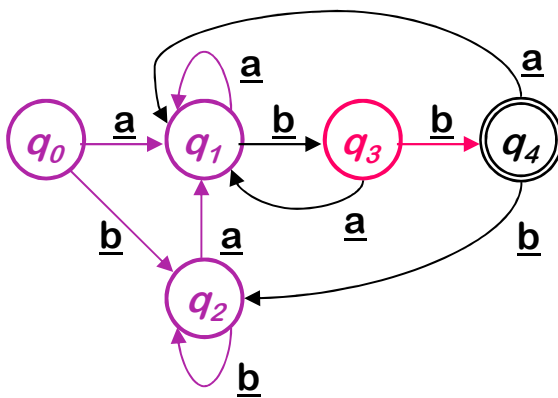
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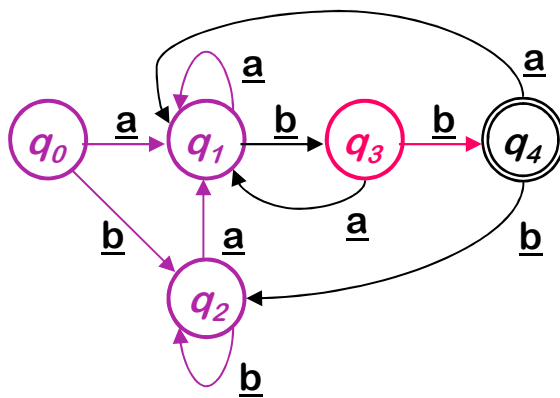
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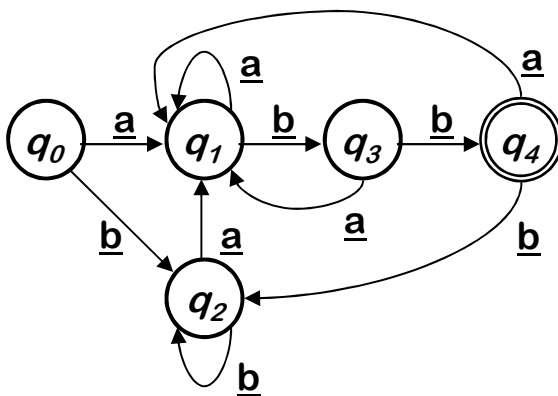
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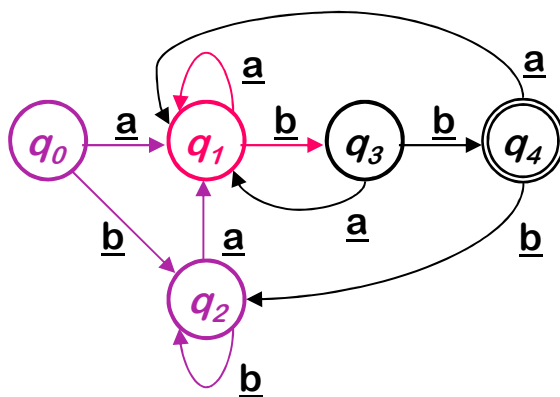
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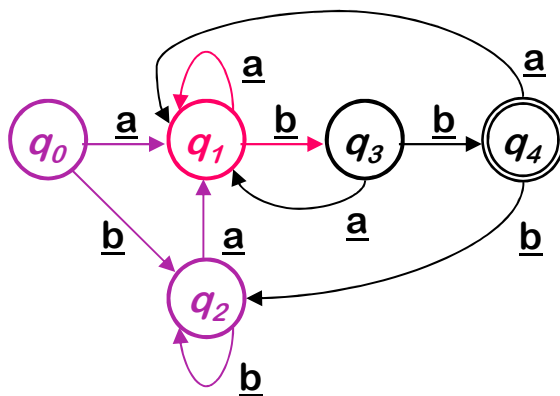
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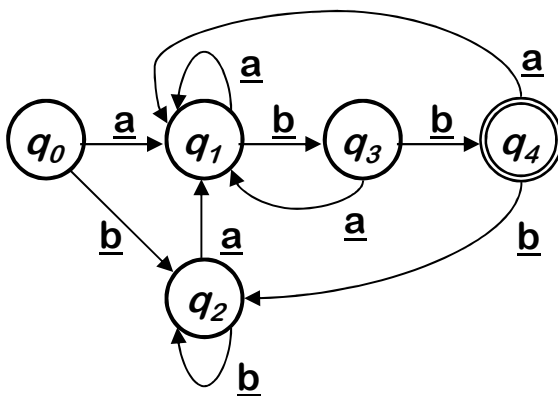
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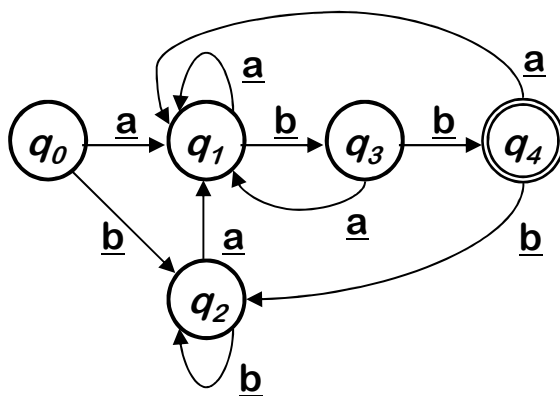
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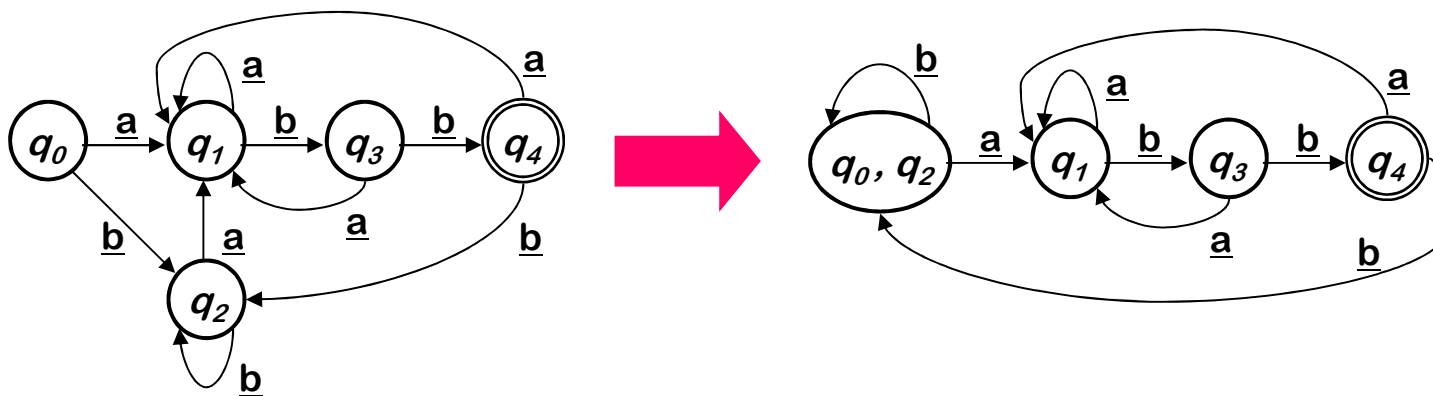


Empty worklist $\Rightarrow$ done!

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20% reduction in number of states 15